

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EE) Examination

November 2012

EE205 Electrical Machines - I

Date: 05.11.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Explain the features of electro mechanical energy conversion. Also explain the energy balance equation. [05]
- (b) Explain the energy and coenergy with help of necessary diagram. [02]
- Q - 2 (a) Explain the different types of d.c. generators. [04]
- (b) Explain the losses in a d.c. machine and draw the power flow diagram. Derive the condition for maximum efficiency. [06]
- (c) The armature of a d.c. generator consists of 40 coils and each coil has 20 turns. When the armature is rotated at 200 radians per second in a 4 pole field structure having a flux of 5 mWb/pole and there are four paths in the armature, calculate (i) number of armature conductors (ii) the voltage between brushes generated by the armature. [04]

OR

- Q - 2 (a) A short shunt compound generator supplies a current of 100 A at 220 V. The resistance of the shunt field is $50\ \Omega$, of the series winding is $0.025\ \Omega$ and of the armature $0.05\ \Omega$. Iron and friction losses amount to 1 kW. Find (i) the e.m.f. generated (ii) the copper losses (iii) the b.h.p. of the prime mover (iv) commercial efficiency. [07]
- (b) Define the term armature reaction with respect to d.c. machines. Explain in detail the armature reaction. Write the equation for calculation of demagnetizing ampere turn per pole, cross-magnetizing ampere turn per pole and ampere turn per pole for compensating winding. [07]
- Q - 3 (a) A d.c. series motor takes 20 A at 100 V and runs at 950 r.p.m. Armature resistance is $0.05\ \Omega$ and series field resistance is $0.2\ \Omega$. At what speed will it run developing same torque but with a $10\ \Omega$ resistance placed in parallel with the armature? Assume that the flux is proportional to the field current. [07]

- (b) Draw and explain the following characteristics for d.c. series motor (i) Torque versus armature current, (ii) Speed versus armature current, (iii) Speed versus torque. [07]

OR

- Q - 3 (a) Explain the different type of electric braking methods used for d.c. motors. [04]
(b) Explain the various methods used for speed control of d.c. shunt motors. [06]
(c) Explain the necessity of starter in d.c. motor. Differentiate between three point starter and four point starter. [04]

SECTION – II

- Q - 4 Define the term transformer. Explain the operating principle of transformer. Derive the e.m.f. equation of a transformer and also give the classification of transformer. [07]
- Q - 5 (a) Draw the exact equivalent circuit of a loaded transformer and explain the significance of the parameter used in equivalent circuit. [07]
(b) A 150 kVA transformer is loaded as under: Load increases from zero to 100 kVA in 3 hours from 7 a.m. to 10 a.m., stays at 100 kVA from 10 a.m to 6 p.m. and then the transformer is on no load till next day. Assuming the load to be resistive and core loss equal to full load copper loss of 1 kW, determine the commercial and all-day efficiency of transformer. [07]

OR

- Q - 5 (a) A 80 kVA, 2000/200 V, 50 Hz single phase transformer has impedance drop of 8% and resistance drop of 4 %. Calculate the regulation of the transformer at full load 0.8 p.f. lagging. Also find the power factor at which voltage regulation will be zero. [07]
(b) Explain the procedure for open circuit and short circuit test on transformer and explain the advantages of transformer tests. [07]
- Q - 6 (a) Write a technical note on auto transformer. [07]
(b) Explain the condition for satisfactory parallel operation of two three phase transformer. [07]

OR

- Q - 6 (a) Draw and label all the connection diagram of three phase transformer connection. [07]
(b) A three phase step down transformer is connected to 6600 V on the primary side. The ratio of turns per phase is 12 and the line current drawn from the mains is 20 A. Find the secondary line voltage and secondary line current if the transformer is connected in (i) star-star, (ii) delta-delta, (iii) star-delta, (iv) delta-star. Neglect losses. [07]
